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- Since $\sum_i a_i = \sum_j d_j$
  if $\mathbf{x} = (x_{ij})_{mn \times 1}$ satisfies any $(m + n - 1)$ equations then it automatically satisfies all the $(m + n)$ equations.

- The constraints are of the form
  $A\mathbf{x} = \mathbf{b}$,
  where

$$A_{(m+n)\times mn} = \begin{bmatrix}
\underset{n}{\overbrace{111..11}} & \mathbf{0_n} & \mathbf{0_n} & \dots & . & \mathbf{0_n} \\
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\mathbf{0_n} & \mathbf{0_n} & \underset{n}{\overbrace{111..11}} & \mathbf{0_n} & . & \mathbf{0_n} \\
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\overbrace{100...0} & \overbrace{100...0} & . & . & \dots & \overbrace{100...0} \\
\overbrace{010...0} & \overbrace{010...0} & . & . & \dots & \overbrace{010...0} \\
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- D is any nonsingular sub matrix of A if and only if D has the same structure as **B**.

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- In a transportation array (**TA**) each cell corresponds to a variable, that is the (i, j)th cell corresponds to variable x_{ij}.

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- Any set of ≤ 2 cells is LI.
- Any set of $\geq m + n$ cells is LD.

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i.
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- **Definition 2:** A subset of (m+n-1) **cells** of the **TA** is said to be a basic set if they are linearly independent. The cells in a basic set are called basic cells.

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- Since we are now solving $\mathbf{B}x_B = [a_1, .., a_m, d_1, .., d_{n-1}]^T$ and each row of **B** corresponds to a constraint (supply or demand).
- There exists a constraint in the **TP** which has **exactly one** basic variable corresponding to **B**.

- Since each **row** and **column** of the transportation array corresponds to a **constraint**, there exists a row or column of the transportation array which has **exactly one** cell from the basic set $\mathcal{B}$.

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- If $\mathcal{B}$ is a **basic** set of cells and if the **row or column** having a single basic cell is struck off from the **TA**, then in the **reduced** (or remaining) array there will again be a **row** or **column** with a **single** basic cell.

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- If $\mathcal{B}$ is a **basic** set of cells and if the **row or column** having a single basic cell is struck off from the **TA**, then in the **reduced** (or remaining) array there will again be a **row** or **column** with a **single** basic cell.
- Since every row and column of the array has at least one basic cell, one can continue till **all** the rows and columns of the **TA** are **struck off**.

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	$j=1$	2	3	4	5	6	a_i
$i=1$							7
2							17
3							5
4							24
d_j	15	10	9	3	8	8	

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- Let us first start with cell $(2,3)$ as a basic cell and then try to construct a **BFS** of the above problem.

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- Let us first start with cell $(2,3)$ as a basic cell and then try to construct a **BFS** of the above problem.
- Since the minimum of a_2 and d_3 is $d_3 = 9$, we take $x_{23} = 9$. Delete the third column and change a_2 from 17 to $a_2' = 17 - 9 = 8$.

- In the reduced array choose a basic cell say $(2, 4)$. Take $x_{24} = 3$ since $3 = min\{d_4 = 3, a_2' = 8\}$. Proceeding in this way we get the following **BFS**.

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	$j=1$	2	3	4	5	6	a_i
$i=1$		[7]					7
2			[9]	[3]	[5]		17
3					[3]	[2]	5
4	[15]	[3]				[6]	24
d_j	15	10	9	3	8	8	

North West Corner Rule for Initial BFS

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-

					a_i
	40				40
					30
					60
d_j	50	20	40	20	

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North West Corner Rule

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- Start by choosing a basic cell from the North-West corner of the array and assign the maximum possible value (maintaining feasibility) to that variable.
-

					a_i
	40				40
	10	20			30
					60
d_j	50	20	40	20	

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-

					a_i
	40				40
	10	20			30
					60
d_j	50	20	40	20	

- Delete the first row of the TA (the constraint being taken care of) and repeat the above process with the North-West corner of the reduced array.

North West Corner Rule

					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

North West Corner Rule

-

					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

- Delete the first column of the array and repeat the above process by starting the North-West corner of the reduced array.

North West Corner Rule

					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

- Delete the first column of the array and repeat the above process by starting the North-West corner of the reduced array. The following tables shows the various steps following the North-West corner rule till a **BFS** is obtained.

North West Corner Rule

North West Corner Rule

-

					a_i
	40				40
	10	20			30
		0	40		60
d_j	50	20	40	20	

North West Corner Rule

-

					a_i
	40				40
	10	20			30
		0	40		60
d_j	50	20	40	20	

-

					a_i
	40				40
	10	20			30
		0	40	20	60
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θ -loops

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  1. Nonempty.

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  3. No proper subset of this collection satisfies both property **1** and property **2**.

- Consider the following examples.
-

	1	2	3	4
1	$\circ$	$\circ$		
2	$\circ$	$\circ$		
3				
4				

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	1	2	3	4
1	$\circ$	$\circ$		
2	$\circ$	$\circ$		
3				
4				

-

	1	2	3	4
1	$\circ$			$\circ$
2	$\circ$	$\circ$		
3		$\circ$	$\circ$	
4			$\circ$	$\circ$

-

	1	2	3	4
1	○			○
2	○	○		
3		○		
4				

-

	1	2	3	4
1	○			○
2	○	○		
3		○		
4				

-

	1	2	3	4
1	○	○		
2	○	○		
3			○	○
4			○	○

	1	2	3	4
1	○			○
2	○	○		
3		○		
4				

	1	2	3	4
1	○	○		
2	○	○		
3			○	○
4			○	○

- In the third and fourth example, the marked cells do not form a θ- loop of the 4×4 transportation array, since it violates properties 2 and 3, respectively.

-

	1	2	3	4
1	∘			∘
2	∘	∘		
3		∘		
4				

-

	1	2	3	4
1	∘	∘		
2	∘	∘		
3			∘	∘
4			∘	∘

- In the third and fourth example, the marked cells do not form a θ- loop of the 4×4 transportation array, since it violates properties 2 and 3, respectively.
- The first and the second however are θ- loops.

- **Theorem 4:** The cells in a θ- loop are **LD**.

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  Theorem 6 : If $\triangle \neq \phi$ is a collection of cells of the **TA** which contains no θ- loop, then $\triangle$ is **LI**.

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  Corollary 6: A subset of cells $\triangle$ of the **TA** is **LI** if and only if it contains **no** θ- loop.

- **Theorem 7:** If $\mathcal{B}$ is a collection of $m + n - 1$ basic cells of the **TA** and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ-loop and this loop includes the cell (p, q).

- **Theorem 7:** If $\mathcal{B}$ is a collection of $m+n-1$ basic cells of the **TA** and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ-loop and this loop includes the cell (p, q).
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- **How to get the optimal solution from a given basic feasible solution?**
  Let $\mathbf{x} = (x_{ij})$ be the initial BFS.
- The dual of the **TP** is given by
  Max$\sum_{i=1}^{m} a_i u_i + \sum_{j=1}^{n} b_j v_j$
  subject to,
  $u_i + v_j \leq c_{ij}$ for all $i = 1, .., m$, $j = 1, .., n$.

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- This set of equations is obtained from $\mathbf{y}^T B = \mathbf{c}_B^T$, where $\mathbf{y}^T = [u_1, \ldots, u_m, v_1, \ldots, v_{n-1}]$.
- We have $m + n - 1$ equations and $m + n - 1$ unknowns, which can be easily solved.

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- **Step 3:** Find the θ- loop in $\mathcal{B} \cup \{(p, q)\}$, where the cell (p, q) is such that
  $c_{p,q} - u_p - v_q = min\{c_{ij} - u_i - v_j : c_{ij} - u_i - v_j < 0\}$.

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- **Step 4:** Assign value $+\theta$ to cell (p,q) and alternately assign $+\theta$ and $-\theta$ to all the cells in the θ- loop, so that the sum of the allocations ($+\theta$ and $-\theta$ allocations) in each row and column add up to zero.

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- Take $+\theta = min\{x_{ij} \in \theta\text{-loop} \ : \ \text{cell } (i,j) \text{ is assigned value } -\theta\}$. Find the new BFS say $\mathbf{x}'$ where x'_{ij} is either equal to x_{ij}, $x_{ij} + \theta$ or $x_{ij} - \theta$.

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- If x_{pq} is a non basic variable in a **BFS** then the corresponding column in the simplex table $B^{-1}\tilde{\mathbf{a}}_{p,q} = \mathbf{u_{pq}}$, is such that $u_{\mathbf{k},pq} = -1, 1,$ or 0

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- If x_{pq} is a non basic variable in a **BFS** then the corresponding column in the simplex table $B^{-1}\tilde{\mathbf{a}}_{p,q} = \mathbf{u_{pq}}$, is such that $u_{\mathbf{k},pq} = -1, 1,$ or 0 depending on whether the **k** th basic variable gets the allocation θ, $-\theta$ or is not included in the θ-loop in $\mathcal{B} \cup \{(p, q)\}$, respectively.

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- **Example:** Consider the following transportation problem (P) with c_{ij}'s, a_i's (40,30,30) and d_j's (30,50,20) as given below:

2	5	1	40
1	4	5	30
1	5	3	30
30	50	20	

- If x_{pq} (or cell (p,q)) is the **entering variable** of the new basis then according to the minimum ratio rule of simplex, the **leaving variable** is (r,s) if
  $x_{rs} = min\{x_{ij} \in \theta\text{-loop} : \text{ cell } (i,j) \text{ is assigned value } -\theta\}$.
- **Example:** Consider the following transportation problem (P) with c_{ij}'s, a_i's (40,30,30) and d_j's (30,50,20) as given below:

	2	5	1	40
	1	4	5	30
	1	5	3	30
	30	50	20	

- Check whether the initial basic feasible solution $\mathbf{x}_0$ with basic cells
  $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$, is optimal for (P) (by taking $v_2 = 0$, where v_2 is the dual variable corresponding to the second demand constraint).
  Also find the optimal solution.

- The BFS with $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$ as the basic cells is given by
$x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.

- The BFS with $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$ as the basic cells is given by $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:

- The BFS with $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$ as the basic cells is given by $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.

- The BFS with $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$ as the basic cells is given by
  $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- On solving we get, $u_1 = 5, v_1 = -3, u_2 = 4, v_3 = 1, u_3 = 5$.

- The BFS with $\mathcal{B} = \{(1,1),(1,2),(2,2),(2,3),(3,2)\}$ as the basic cells is given by $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- On solving we get, $u_1 = 5, v_1 = -3, u_2 = 4, v_3 = 1, u_3 = 5$.
- The following table shows the $c_{ij} - u_i - v_j$ values against each cell, where we have taken $v_2 = 0$ for easier calculations.

- Check that

- Check that
  $c_{13}-u_1-v_3 = 1-5-1 = -5, c_{21}-u_2-v_1 = 1-4-(-3) = 0,$
  $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3,$

- Check that $c_{13}-u_1-v_3 = 1-5-1 = -5$, $c_{21}-u_2-v_1 = 1-4-(-3) = 0$, $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$, $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:
-

	0	0	-5	40
	0	0	0	30
	-1	0	-3	30
	30	50	20	

- Check that
  $c_{13}-u_1-v_3 = 1-5-1 = -5$, $c_{21}-u_2-v_1 = 1-4-(-3) = 0$,
  $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:
-

	0	0	-5	40
	0	0	0	30
	-1	0	-3	30
	30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.

- Check that
  $c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5, c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0,$
  $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:
-

	0	0	-5	40
	0	0	0	30
	-1	0	-3	30
	30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.
- The most negative value of $c_{ij} - u_i - v_j$ is in cell $(1, 3)$, so this will be the entering variable of the new BFS.

- Check that
  $c_{13}-u_1-v_3 = 1-5-1 = -5$, $c_{21}-u_2-v_1 = 1-4-(-3) = 0$,
  $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:
-

	0	0	-5	40
	0	0	0	30
	-1	0	-3	30
	30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.
- The most negative value of $c_{ij} - u_i - v_j$ is in cell $(1,3)$, so this will be the entering variable of the new BFS.
- The unique θ- loop in $\mathcal{B} \cup \{(1,3)\}$ is given by $\{(1,2),(2,2),(2,3),(1,3)\}$.

- Since $(1,3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1,3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.

- Since $(1,3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1,3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 2)\}$ and the values of the basic variables are given by: $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.

- Since $(1,3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1,3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $\mathcal{B} = \{(1,1),(1,3),(2,2),(2,3),(3,2)\}$ and the values of the basic variables are given by:
  $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,

- Since $(1,3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1,3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $\mathcal{B} = \{(1,1),(1,3),(2,2),(2,3),(3,2)\}$ and the values of the basic variables are given by:
  $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30.$
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,
- $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5.$
  we get

- Since $(1,3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1,3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $\mathcal{B} = \{(1,1),(1,3),(2,2),(2,3),(3,2)\}$ and the values of the basic variables are given by:
  $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30.$
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,
- $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
  we get
- $v_1 = 2, v_2 = 0, v_3 = 1, u_2 = 4, u_3 = 5.$

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
  $\mathcal{B} = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
  $\mathcal{B} = \{(1,1), (1,3), (2,3), (2,2), (3,2)\}$.
-

0	5	0	40
−5	0	0	30
−6	0	−3	30
30	50	20	

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
  $\mathcal{B} = \{(1,1),(1,3),(2,3),(2,2),(3,2)\}$.
-

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31}.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
  $\mathcal{B} = \{(1,1),(1,3),(2,3),(2,2),(3,2)\}$.
-

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31}.
- The unique θ- loop in $\mathcal{B} \cup \{(3,1)\}$ is given by $\{(3,1),(3,2),(2,2),(2,3),(1,3),(1,1)\}$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$, $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$, $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,3),(2,2),(3,2)\}$.
-

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31}.
- The unique θ- loop in $\mathcal{B} \cup \{(3,1)\}$ is given by $\{(3,1),(3,2),(2,2),(2,3),(1,3),(1,1)\}$.
- $(3,1)$ is the entering variable, so if we give $+\theta$ allocation to cell $(3,1)$ (or value of $x_{31} = +\theta$) then $x_{11} = 30 - \theta$, $x_{13} = 10 + \theta$, $x_{23} = 10 - \theta$, $x_{22} = 20 + \theta$, $x_{32} = 30 - \theta$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5, c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5,$
  $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6,$
  $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3.$
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,3),(2,2),(3,2)\}$.
-

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31}.
- The unique θ- loop in $\mathcal{B} \cup \{(3,1)\}$ is given by $\{(3,1),(3,2),(2,2),(2,3),(1,3),(1,1)\}$.
- $(3,1)$ is the entering variable, so if we give $+\theta$ allocation to cell $(3,1)$ (or value of $x_{31} = +\theta$) then $x_{11} = 30 - \theta$, $x_{13} = 10 + \theta$, $x_{23} = 10 - \theta$, $x_{22} = 20 + \theta$, $x_{32} = 30 - \theta$.
- So $\theta = 10$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:

  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:

  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- Check that $v_1 = 2, v_2 = 6, v_3 = 1, u_2 = -2, u_3 = -1$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- Check that $v_1 = 2, v_2 = 6, v_3 = 1, u_2 = -2, u_3 = -1$.
- Check that $c_{23} - u_2 - v_3 = 5 - (-2) - 1 = 6, c_{21} - u_2 - v_1 = 1 - (-2) - 2 = 1$, $c_{12} - u_1 - v_2 = 5 - 0 - 6 = -1$, $c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
$\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

0	-1	0	40
1	0	6	30
0	0	3	30
30	50	20	

- The entering variable is x_{12}.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12}.
- The θ-loop is $\{(3,1),(3,2),(1,2),(1,1)\}$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12}.
- The θ-loop is $\{(3,1),(3,2),(1,2),(1,1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12}.
- The θ-loop is $\{(3,1),(3,2),(1,2),(1,1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12}.
- The θ-loop is $\{(3,1),(3,2),(1,2),(1,1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.
  Any one of x_{11} or x_{32} can be the leaving variable.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12}.
- The θ-loop is $\{(3,1),(3,2),(1,2),(1,1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.
  Any one of x_{11} or x_{32} can be the leaving variable.
- Let x_{32} leave the basis.

- If we take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$.
  we get

- If we take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$.
  we get
- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$

- If we take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$.
  we get
- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1$.
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0, c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3$.

- If we take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$.
  we get
- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1$.
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0, c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3$.
- Since $c_{ij} - u_i - v_j \geq 0$ for all i, j,
  the above BFS is optimal and the optimal value is given by:

- If we take $u_1 = 0$, then by solving the 5 equations given below:
  $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
  $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
  $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
  $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
  $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$.
  we get
- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0$, $c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1$, $c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3.$
- Since $c_{ij} - u_i - v_j \geq 0$ for all i, j,
  the above BFS is optimal and the optimal value is given by:
  $c_{11}x_{11} + c_{12}x_{12} + c_{13}x_{13} + c_{22}x_{22} + c_{31}x_{31} = 2 \times 0 + 5 \times 20 + 1 \times 20 + 4 \times 30 + 1 \times 30 = 270.$